

## Practice Assignment 3

# PRACTICAL EXERCISE: USING SENSORS IN REACT NATIVE WITH TURBO MODULES

## Objectives

- Understand how to use **Turbo Native Modules** to access hardware on Android.
- Practice programming with **NFC (Near Field Communication)** and **Compass (Magnetometer)** in React Native.
- Build an intuitive interface to display sensor data.



Screen 1

## Exercise 1: Using NFC to Read Data from an NFC Tag

### Requirements:

Develop a React Native application that can read data from an NFC tag using **Turbo Native Module** on Android. When the user places an NFC tag near the phone, the tag's information should be displayed on the screen.

### Implementation Steps:

1. **Create a Turbo Native Module to access NFC on Android.**
2. **Register an NFC event listener when the app starts.**
3. **Display the retrieved NFC data on the screen.**

**Outcome:**

- When the user taps an NFC tag on the phone, the tag's information should appear on the screen.
  - The application should use **Turbo Native Module** instead of third-party libraries like `react-native-nfc-manager`.
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## **Exercise 2: Using a Compass to Determine Direction (Screen 1)**

**Requirements:**

Develop a React Native application that uses the **Magnetometer sensor** to determine the device's orientation relative to the North direction. The app should display a **compass interface where the needle rotates based on the device's direction.**

**Implementation Steps:**

1. **Create a Turbo Native Module to read data from the magnetometer sensor.**
2. **Calculate the direction angle based on sensor data.**
3. **Display a compass interface with a rotating needle.**

**Outcome:**

- As the user rotates their phone, the compass needle should dynamically adjust to point North.
  - The application should use **Turbo Native Module** instead of third-party libraries like `expo-sensors`.
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## **Summary**

- This practical exercise helps students understand how to use **Turbo Native Modules** to access hardware in React Native.
- **Task 1** teaches students how to work with **NFC** to read data from tags.
- **Task 2** teaches students how to use the **Magnetometer sensor** to build a dynamic compass.
- Both tasks focus on **performance and native optimization**, without relying on third-party libraries.

**By the end of this exercise, students will be able to build applications that leverage native sensors on Android efficiently!**